

Integrating user-generated content to track urban flooding hotspots and foster emergency management: A case study in central China

Zhongping Zeng^{1,2}  | Jinyu Lan¹ | Jiang Li³ | Qingsong He¹  |
Jiangwei Wang¹ | Siqi Li¹ | Yao Huang⁴

¹College of Public Administration, Huazhong University of Science and Technology, Wuhan, China

²Decision Theater at College of Public Administration, Huazhong University of Science and Technology, Wuhan, China

³Information Center, Department of Natural Resources of Hubei Province, Wuhan, China

⁴School of Software Engineering, Huazhong University of Science and Technology, Wuhan, China

Correspondence

Jiang Li, Information Center, Department of Natural Resources of Hubei Province, Gongzheng Road 27, Wuhan, Hubei, China.
Email: johnlee1124@126.com

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Abstract

This study explores the potential for employing user-generated content (UGC) during severe flooding to discover and track urban flooding disaster hotspots in a timely manner. A flooding case in central China was selected for this study. Crawlers, natural language processing, and geographic visualization methods were used to extract flood-related UGC specific to severely flooded areas. Further, the quality of the geo-tagged content on the web was verified using scientific gauge data (e.g., water level, digital elevation model, and rainfall). Based on our findings, we were able to deduce that UGC on the web is valuable for identifying flooding hotspots. In fact, this approach offers substantial advantages for addressing the emerging needs of data acquisition for flood emergency management. Notably, the values of urban stakeholders that share their observations on the web can be mined rapidly through the integration of various approaches. Overall, the findings of our study can contribute to the efficient mining of web data sources and development of disaster hotspot mapping systems at an urban scale to improve flood management and mitigation. The limitations of UGC and our study's future direction are also discussed.

1 | INTRODUCTION

Urban flooding is one of the most prevalent natural disasters that hinder sustainable development in flood-affected cities. Floods can cause massive economic losses, deaths, and poverty, and even aggravate complex social and ecological issues (World Bank, 2010). According to statistics presented by the United Nations, the economic loss caused by global urban floods amounts to hundreds of billions of dollars (Kunapo, Chandra, & Peterson, 2009; Lo, Xu, Chan, & Su, 2016; World Meteorological Organization, 2008; Zhou, Leng, Su, & Ren, 2019). In the future climate change scenario, it has been predicted that flooding may increase in cities worldwide (Hallegatte, Green, Nicholls, & Corfee-Morlot, 2013). Frequent flooding imposes a series of challenges and pressures on the resilience management of any city. To provide timely rescue services in severely affected areas during the disaster, the ability to cope with urban flooding needs to be improved urgently (Der Sarkissian, Zaninetti, & Abdallah, 2019; Parvin, Surjan, Atta-ur-Rahman, & Shaw, 2016).

Flooding disasters are complex processes with uneven heterogeneity at the spatial and temporal scales (Fugate, 2015). To effectively manage emergencies and disaster relief, decision-makers prefer to use geo-visualization methods, such as disaster hotspot mapping (DHM), to aid in the identification and dynamic tracking of vulnerable and dangerous areas (Elalem & Pal, 2015). In recent decades several approaches, such as field investigation, historical disaster event records, water gauges, and remote sensing, have been developed to collect data for DHM (Degiorgis et al., 2012; Jafarzadegan & Merwade, 2017; Morelli, Battistini, & Catani, 2014). These methods have proven effective for identifying the distribution of disaster hotspots at global and regional scales (Dilley, Chen, Deichmann, Lerner-Lam, & Arnold, 2005). However, these approaches have limitations. For instance, at a finer scale, field surveys are often very labor-intensive and time-consuming and require a wide coverage (Imran, Ofli, Caragea, & Torralba, 2020). During flooding events, remote sensing data are either unavailable or patchy because flooding areas are often covered by clouds or shadows (Dottori et al., 2016; Li, Wang, Emrich, & Guo, 2018). In fact, even in developed countries, disaster situational awareness is troubled by insufficient water-level monitoring facilities (Wang, Mao, Wang, Rae, & Shaw, 2018). Therefore, urban flooding management needs more channels of data sources to improve the capability of sensing technology with respect to disasters.

Recently, the emergence of citizen science and voluntary geographic information (VGI; humans as sensors) has provided new opportunities for urban disaster management (Goodchild, 2007; Houston et al., 2015; Raza et al., 2020; Stokes & Senkbeil, 2017). User-generated content (UGC) on the web contains abundant information regarding flood location in the form of text, pictures, and videos (Rossi et al., 2018; Weeser et al., 2018). Notably, geo-tagged information aids in flood disaster management, enabling information sharing, risk communication, disaster monitoring, disaster reduction, life rescue, and risk assessment (Hao & Wang, 2020; Kankanamge, Yigitcanlar, Goonetilleke, & Kamruzzaman, 2020; Yang et al., 2019). Traditional search engines can help to quickly retrieve event-related webpages according to specific keywords; however, search engine optimization tools mainly focus on topic-relevant webpages across different websites, instead of webpage content analysis. Previous studies have used UGC (e.g., Twitter) for crime hotspot analysis (Gerber, 2014); however, with respect to disaster management, the value of named entities (e.g., place names, dates, times, and values) on various websites has not been efficiently explored and used (Bikel, Schwartz, & Weischedel, 1999; Gelernter & Balaji, 2013). Additionally, as suggested by previous studies, the quality of UGC for disaster management should be emphasized (Bhuvana & Arul Aram, 2019; Feick & Robertson, 2015; Michelsen et al., 2016).

This study aims to explore the potential for using geographical name information of flooded places scattered across various webpages during a disaster to establish a DHM system. We considered the 2016 rainstorm event in Wuhan, China, as our case study. Based on our results, during emergencies, unstructured UGC supported by GeoAI (e.g., web crawlers, natural language processing (NLP), data mining, and geo-visualization) can be valuable for integration into flooding disaster recognition to enable better decision-making. The intrinsic relations of the geo-referenced information in different webpages are valuable for DHM, according to a comparison of the

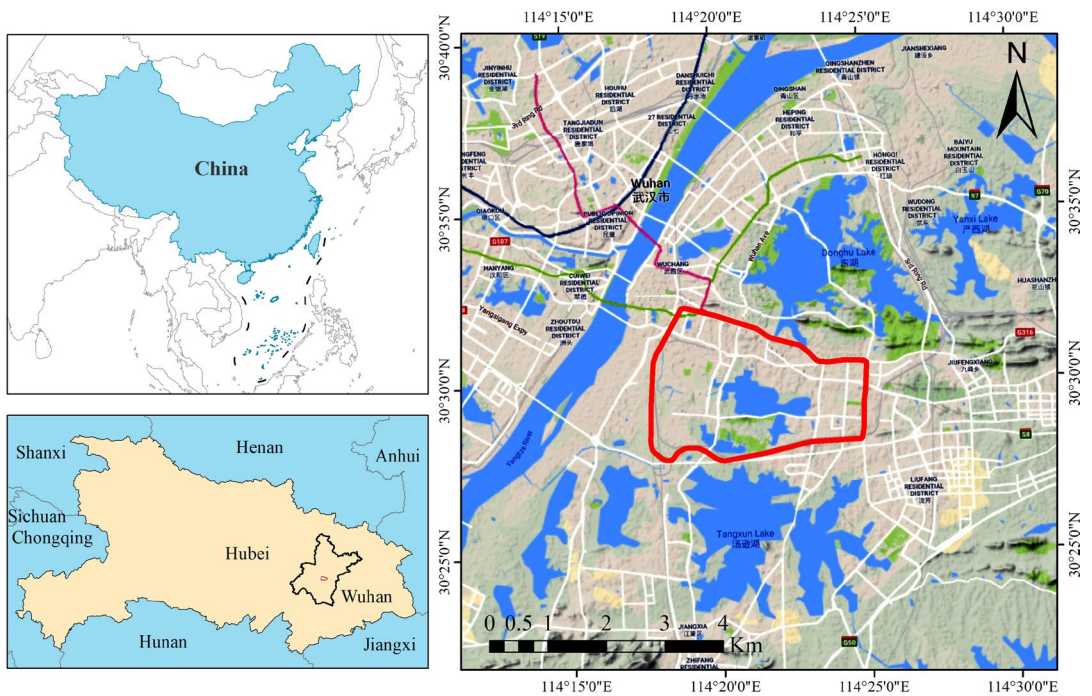


FIGURE 1 The study area: Wuhan, central China

geo-referenced data to rainfall data, digital elevation models (DEMs), and water gauges. Several strategies have also been proposed to overcome the challenges associated with UGC mining for flood emergency management. To our knowledge, this is one of the earliest studies to investigate the value of UGC for DHM in the context of China.

2 | EVENT BACKGROUND

2.1 | Event introduction

The city of Wuhan, which has a population exceeding 10 million and is located along the Yangtze River, is the capital of Hubei Province in central China (Figure 1). With a subtropical monsoon climate, the city receives an average rainfall of ~1,302 mm/year, primarily between April and July. Historically, the city has suffered from flooding hazards caused by heavy rainstorms because of its low-lying plain areas and poor drainage system (Liu, Wang, & Li, 2014); in fact, the heavy rainfall that occurred in the summer of 2016 far exceeded the capacity of the urban drainage system (2016 precipitation, 100 mm/day; capacity of the drainage system, 34.5 mm/hr) (Huaxia, 2013). Therefore, torrential rains in Wuhan led to flooding and, consequently, waterlogging at hundreds of locations; the water levels of Nanhu and other lakes rose rapidly and eventually exceeded those of dams. The communities around the wetlands became victims of the floods. Seven people died, and thousands of livelihoods were severely affected.

The heavy rainfall event in 2016 was ranked as one of the most severe and prolonged flooding disasters in the city over the past decade (Hao, 2016). Although the wet season started at the beginning of June 2016, the most intensive rainfall occurred in the first week of July. The meteorological data indicated that the rainfall reached 932.6 mm from June 1 to July 6, of which 582.3 mm fell between June 30 and July 6. Such data indicate that almost half of the annual average precipitation was received in 7 days (Figure 2). Between 20:00 on July 5

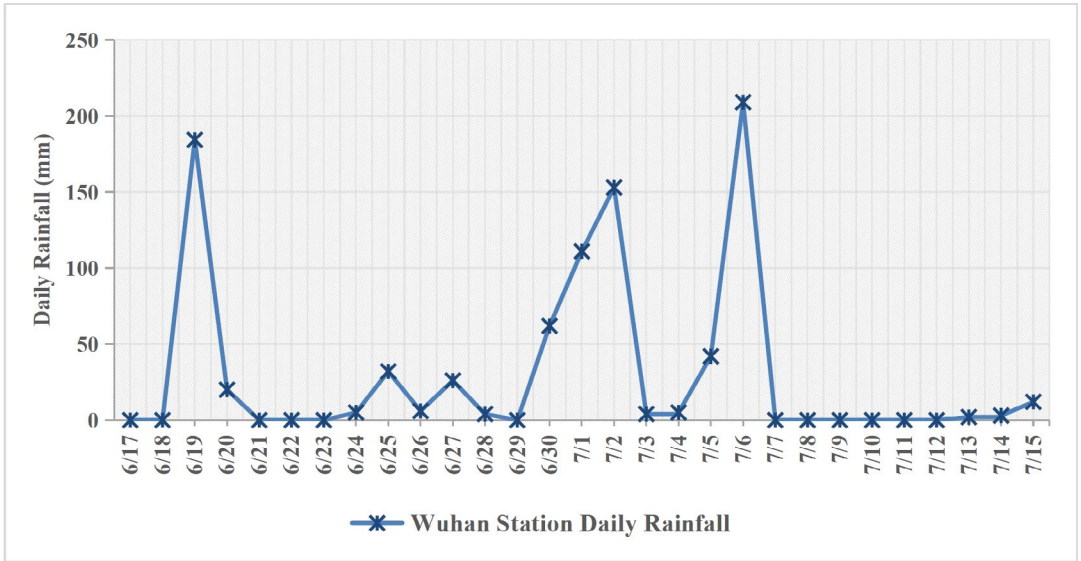


FIGURE 2 Daily rainfall between June 17, 2016 and July 15, 2016 in the study area

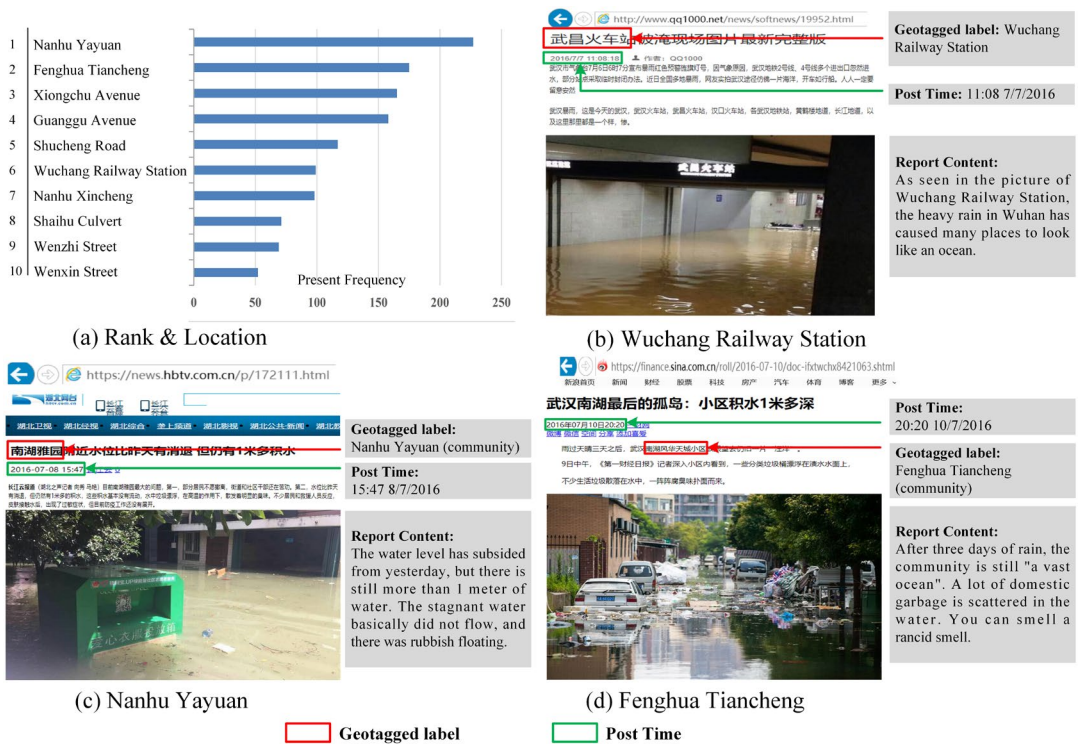
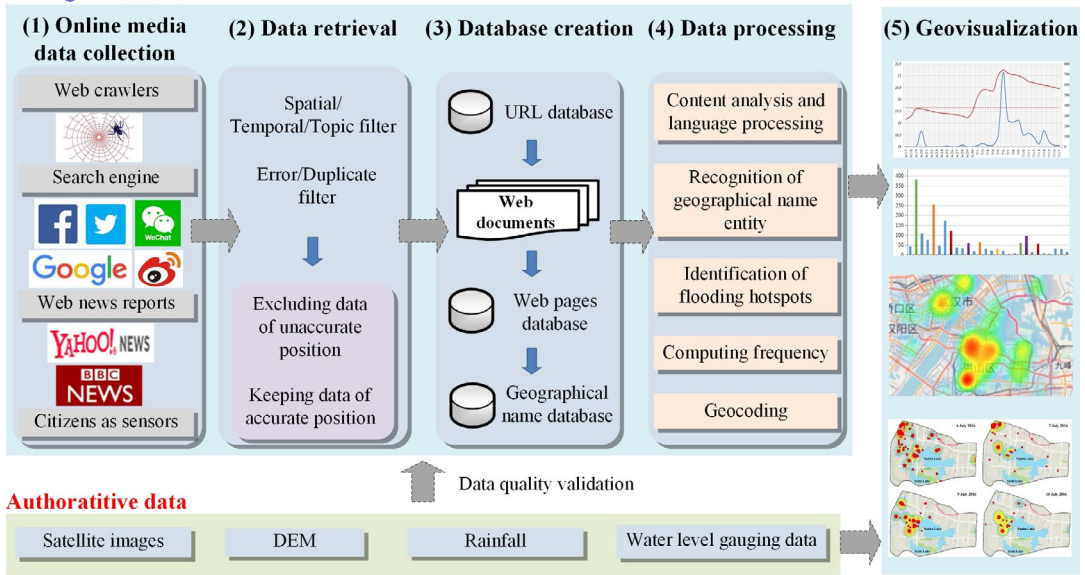


FIGURE 3 Flooding hotspots and their representation in user-generated content

and 13:00 on July 6, rainfall exceeded 150.21 mm, and in some regions of the study area the rainfall reached 241.5 mm. Because of the extreme climate and poor drainage system of the city, flooding and urban waterlogging lasted until July 14.

User generated content

**FIGURE 4** System framework**2.2 | Media data characteristics**

According to several studies, natural disasters provoke widespread discussions on the web (Crooks, Croitoru, Stefanidis, & Radzikowski, 2013). Over the last decade, the internet in China has experienced a rapid transition characterized by "social media transformation" and "media integration" (Huang & Lu, 2017). It is estimated that the number of Chinese internet users exceeded 0.71 billion in 2016. Equipped with smart mobile devices and social network systems, citizens are empowered to record environmental changes and become reporters of disasters online, similar to journalists (Houston et al., 2015). The public sector also leverages various web platforms (e.g., Website, WeChat, and Weibo) to release news. Therefore, widespread UGC on the web provides opportunities to gather relevant information during catastrophic natural disasters.

The disastrous flooding event of 2016 triggered many reports and debates on the web across China (Hao, 2016). The authoritative media used official websites (Figure 3c), public accounts, bulletin board systems, and blogs to report and track disaster events (e.g., CCTV, Xinhua Net, Phoenix Net, and central and local government websites). Individuals and businesses also used WeChat official accounts, Zhihu (similar to Quora), Sina Blog, Tianya communities (a bulletin board service), Baidu Post Bar, Baidu Baike (similar to Wikipedia), and other websites to publish and exchange disaster information. The media outlets were dominated by witness reports (texts, photos, and videos) about communities, apartments, and streets blocked by water (Figures 3b–d). In particular, areas with severe disasters and high vulnerability received concentrated attention from the media and the public.

Usually, search engines can help the public obtain pictures and text of disaster information from the web. This type of "big data" usually has the following characteristics. First, the data vary in structure, have high velocity, and are supplied in large quantities, which increases the challenge of obtaining valuable information. Second, the data exist in overwhelming quantities and are difficult for humans to process; manual processing would require significant time, which cannot meet the requirements imposed by real-time dynamic decision-making (Guo et al., 2020; Hao & Wang, 2020; Mohanty et al., 2021; Stock, 2018). Finally, traditional search engines might not explore the spatial relations among different geo-tagged information across different webpages. For instance, the associations between flooding locations are found in sentences with different web content. The exploration of such spatial associations might help us gain a deeper understanding of the geographical phenomena, causes, and process of

a disaster. Therefore, there is a need to develop specific tools to extract information related to flood disaster locations.

3 | METHODS

Identifying geographic information sourced from webpages is a complex task. In this study, a crawler framework was designed to retrieve flooding locations from UGC. The framework is shown in Figure 4 and contains several different parts: UGC data collection and retrieval; database creation (for the storage of the linkage index, web contents, and flooding locations); data processing (UGC content analysis based on NLP); and geo-visualization. The Python programming language and Google Maps services were employed to implement a system prototype for the dynamic monitoring of flooding locations.

3.1 | Data collection and retrieval

The crawler offers a mechanism to acquire and compile web content batches. To train the crawler to identify webpages related to the event, a keyword list was created. These keywords consisted of single words (and their combinations) related to the subject of the event (e.g., floods, rainstorms, and waterlogging disasters), the location of the event (e.g., Wuhan and Nanhu), and the time attributes of the event (e.g., the time of the event, 2016, and the release time). For example, by using “Wuhan”, “heavy rain”, “flood”, “2016”, “Nanhu”, “rescue” and their combinations, the crawler could quickly retrieve reports related to the event. To expand the scope of media coverage as much as possible, while improving search efficiency, multi-threading methods were used to connect different search engine tools to help collect the flood data. Every report link and content that might be related to the event was recorded in the database after manual deduplication checks.

3.2 | Database creation

The database contained six tables: (1) user information; (2) query keyword; (3) search log; (4) URL record; (5) look-up; and (6) geographic disaster location. The user information table was used to store the user information inputted by the system. The query keyword table recorded all query keywords used by the login user. The search log table stored the URL address, disaster event identity, and query keyword identity. The look-up table contained information on the place names. The disaster location table recorded the disaster locations and names of flooded places (longitude, latitude, and frequency). Each location name found on the webpages was recorded with the sentences and web linkages to ensure that the retrieval results could be traced to their URL link through the association relationship of the database for analyzing and validating the information.

3.3 | Data processing based on natural language processing

Geographical information retrieval (GIR) from webpages still remains a challenge. NLP is an effective method that can aid in the identification of named entities using word segmentation, which is widely used in information retrieval, information extraction and filtering, and text clustering analysis (Collobert & Weston, 2008; Devlin, Chang, Lee, & Toutanova, 2018; Mikolov, Chen, Corrado, & Dean, 2013). NLP is useful for identifying information related to geographical names and simplifying this process (Steedman, 1996). In our study, we used a Chinese application called “Jieba Word Segmentation” to enable quick processing of the geographical name entity.

Several methods based on statistical machine learning (e.g., naive Bayes, decision trees, hidden Markov models, maximum entropy, and look-up tables) and deep learning (e.g., convolutional neural network and recurrent neural network) algorithms can be employed to enhance the accuracy and efficiency of syntactic and semantic analysis of sentences (Kim, Park, & Suh, 2020; Laylavi, Rajabifard, & Kalantari, 2017; Luo, 2021; Luo & Chen, 2020; Zhang et al., 2011). The main aim of our study was to investigate the potential of UGC for DHM. We used the "look-up table" method described by Ghasemi and Licayan (2013) for place name recognition. This method requires that the name of the place of the disaster be known in advance.

In the Chinese context, word formation and semantic representation of place names often follow certain common characteristics. For instance, a place name has a special address suffix name ("街" street, "路" road, "站" station, "社区" community, etc.). Changes in the words before the suffix name indicate different places. The same

```
//pseudo-code for the algorithm to extract the place names

//function to get common string of place names from different sentences

//which contain suffix //"街(street) and 路(road)"

String GetPlaceNameString (Sentence1, Sentence2, String Suffix) {

//identify the start position of suffix in the sentences

Sentence1_startposition = Sentence1.indexOf (Suffix);

Sentence2_startposition = Sentence2.indexOf (Suffix);

//do loop: in the start position, the preceding words of the suffix name in the different
//sentences were iteratively extracted and were compared, until the preceding words
//appeared to be different.

While (Sentence1_startposition >=0&& Sentence2_startposition >=0&&

Sentence1.charAt(Sentence1_startposition) ==

Sentence2.charAt(Sentence2_startposition))

{

PlaceName= Sentence1.charAt(Sentence1_startposition) + PlaceName;

Sentence1_startposition --;

Sentence2_startposition --;

}

Return PlaceName;

}
```

place has the same preceding words despite being found on different webpages. A preliminary analysis showed that most flooding locations on streets, residential quarters, traffic stations, traffic arteries, squares, wetlands, rivers, and communities received more attention in the event. Based on such language characteristics, we designed an algorithm that could quickly extract sentences containing the specific suffixes of place names. The preceding words of the suffix name were iteratively extracted until the preceding words appeared to be different. Through this approach, the name entities of flooding places were effectively identified. Combined with manual intervention, a look-up table was subsequently constructed. Because some place names will appear in multiple locations in a webpage, the program codes were optimized to ensure that these place names were calculated only once.

Previous studies proposed a manually created gold standard to score the algorithm (Buscaldi & Rosso, 2008; Gelernter & Balaji, 2013; Ghasemi & Licayan, 2013). Two participants were invited to evaluate the accuracy of the geographical names searched by our software. Each was given the same set of UGC and a blank spreadsheet. Given the same instruction, the participants independently completed their coding of the place names (e.g., flooded streets, buildings, and communities) on the blank spreadsheet. The results were compared with the flooding place names retrieved by the crawler system. Cohen's (1960) kappa coefficient analysis was employed to assess the reliability and accuracy of flooding locations retrieved by the crawler.

3.4 | Kernel density estimation

Geographic visualization tools were used to track flood hotspots that changed over temporal and spatial contexts. Kernel density estimation (KDE) is a method that intuitively predicts the unit density of point and line feature values in a continuous region, according to the distribution of discrete values (Elalem & Pal, 2015; Majumder et al., 2019; Mazumdar & Paul, 2018). The kernel function is represented as a bivariate probability density function. If $S_1, \dots, S_n$ are independent and identically distributed samples extracted from the population, then the kernel density at the estimated point s will be:

$$f(s) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h^2} k\left(\frac{s - s_i}{h}\right) \quad (1)$$

where $f(s)$ is the kernel density calculation function at spatial position s ; h is the distance attenuation threshold; k is the kernel function; n is the sample size within the range of the distance scale; and $s - s_i$ represents the distance from the estimated point to the sample s_i . In our study, frequency computing and KDE were employed in combination for hotspot analysis.

3.5 | Data quality assessment method

Goodchild and Li (2012) proposed several approaches (e.g., crowdsourcing, social, and geographic) to address the quality of VGI. In general, scientific gauges (e.g., water level, DEM, and rainfall data) are often used to perform quality assessments to ensure that UGC can be employed in scientific research and practical management. Previous studies compared authoritative data (e.g., rainfall and water-level data) with the frequency of UGC reports (e.g., Twitter posts) to validate the association between the relevance of UGC and the severity of disaster events (de Albuquerque, Herfort, Brenning, & Zipf, 2015). Horita, de Albuquerque, Marchezini, and Mendiondo (2017) applied a model-based framework to identify useful information for disaster decision-making. Geographic knowledge relevant to topographical phenomena (e.g., "water flows downwards") informs us that lower elevation areas are more likely to suffer waterlogging. Thus, authoritative data may help to evaluate the potential of UGC. Based on previous practices, we used rainfall, water-level gauging data, and DEM (obtained from the official website of the US Geological Survey: <https://www.usgs.gov/>) to evaluate the data quality of UGC.

4 | RESULTS

4.1 | Data crawling results

The crawlers required approximately 15 min to retrieve all possible hyperlinks and flooding location names on December 12, 2018, using different terms. The web crawler was used through an Intel® Core™ i5-8250U CPU (1.60 GHz, 8 GB RAM) processor. Compared to the manual mode (which takes 12 days/person), the crawler markedly improved the efficiency of information retrieval from a huge number of webpages. After removing errors and duplicates, 3,855 webpage URLs were retrieved from 2,106 websites. Consistent with previous studies, not all the webpages were relevant to the flooding event (de Albuquerque, Herfort, Brenning, & Zipf, 2015). Finally, 2,430 on-topic URLs were obtained for further analysis.

4.2 | Analysis of the flooding hotspot results

Not every URL manually identified as relevant to the event contained the flood location name. Approximately 420 links and webpages (approximately 10%) were identified as event-relevant. Almost all webpages that contained the

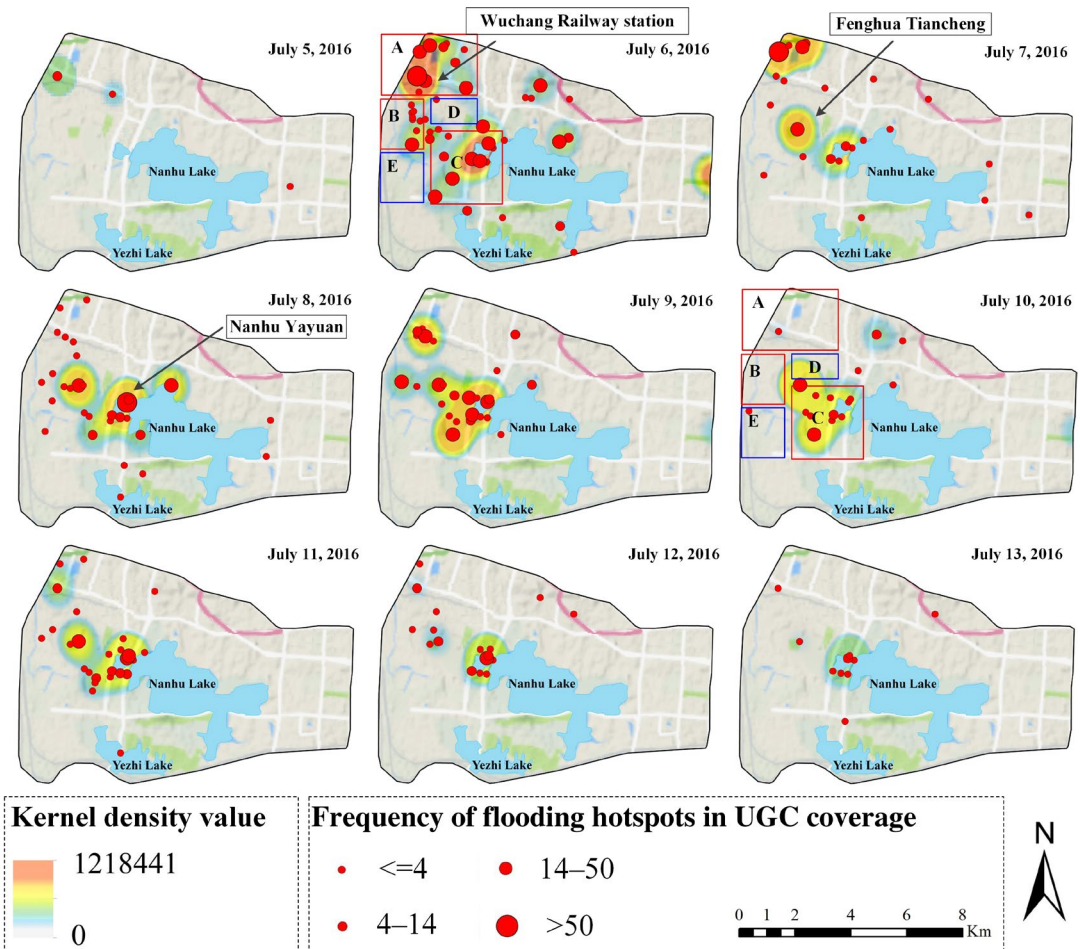


FIGURE 5 Flooding hotspots tracked using daily user-generated content coverage

name of a specific flood location were related to the 2016 flood event. Open map services (e.g., Baidu or Google Maps) were adopted to identify the coordinates of the locations. After checking the accuracy of the method using the gold standard, the results showed that the kappa coefficient exceeded 84%, and there was a very good agreement between the participant's judgment and our software with respect to the recognition of the geographical names of flood areas in the region.

4.3 | Evolution of flooding hotspots

As shown in Figure 5, on July 5, 2016, there were only a few flooded areas. However, due to the continuous rainstorm at midnight on July 5, the water level of Nanhu Lake rapidly rose and water eventually poured into the urban area, with catastrophe consequences. The number of disaster reports dramatically increased, reaching a maximum on July 6. News coverage appeared to decline on July 7. However, to our knowledge, the water level on that day remained high and the disaster situation persisted (Figure 7a), which implied that the harvested UGC did not reflect the real situation of that day. Nonetheless, there was an overall consistency between the reduction of disasters and decreased media reports. According to the water-level data, the number of media reports tended to decline as the flood alleviated.

The flooding did not recede until July 14. According to the results of hotspot mapping, on July 6, at least three districts (zones A, B, and C) should have drawn the attention of emergency sectors. Zone A received the most reports on July 6. The transportation hub of the center of China is located in this area. Thus, if this area suffered severe waterlogging for a long time, it would attract more attention (Figures 3b and 5). From July 6 to July 9, the disasters in zones A and B were alleviated. However, the rescue activities in the two communities named "Nanhu Yayuan" and "Fenghua Tiancheng" (Figures 3c–d and 5) in zone C received continuous interest, which indicated that the zone suffered most during this event. Notably, there were no reports for zones D and E during the rainstorm despite their low altitude. The interpretation of the remote-sensing image indicates that these two zones are farmland and campus playgrounds, respectively. According to the DEM, the above areas might have been flooded but received less attention than the densely populated regions; these areas have lower asset values and may not have been noticed by news media.

Compared to traditional search engines, which usually return a list of webpages without giving extra notice to the flooding location information (Ghasemi & Licayan, 2013), our methods and crawler tools described in Section 3 can extract flooding place names from webpages across different websites. The hotspot map illustrates the spatial-temporal distributions of flooding areas, aiding in the monitoring and reproduction of disaster outbreaks and evolution from a new perspective. Aided by such tools, emergency management groups can share and analyze crisis situations in real time. In addition, such maps can be overlaid with other data sources (e.g., DEMs, water-level gauging data, and risk analysis maps) to improve the quality of decision-making, taking into account that UGC provides additional information during an emergency. The symbols (circle size) in the hotspot maps (Figure 5) represent the relative risk levels. Notably, a dynamic hotspot map sequence visualizing the spatial-temporal evolution pattern of disasters can help experts increase group activity awareness. Valuable information, such as severely affected locations identified by the UGC data mining process, can facilitate the identification of locations that need more support for rescue and emergency decision-making. In addition, the hotspot map can be further developed for the public to freely access via the internet, thereby improving risk communication and promoting the public's hazard risk perception in an intuitive manner.

4.4 | Data quality assessment and validation

Geographic knowledge was employed to examine the potential of using UGC to identify flooding hotspots (Goodchild & Li, 2012). The flooding disaster process triggered by rainfall events followed the sequence: heavy

rain → water accumulated in low-lying areas → water level increasing → causing damage in vulnerable areas (e.g., high-density communities and population, critical infrastructure, and high-density buildings) → users' response and observations → post appears on the web. Thus, rainfall, hydrological data, DEM, and socioeconomic data (e.g., land use and land cover based on remote sensing, and population survey data) can be useful for evaluating the quality of UGC.

4.4.1 | Rainfall

According to the rainfall data, the wet season began on June 1, 2016. From June 17 to July 6, there were five obvious rainfall events occurring on June 19, June 25, June 27, July 1, and July 5 (Figure 2). Rainfall was heaviest on June 19, July 1 and July 5, reaching a new peak on these days (Figure 6a). Extremely short rainfall may trigger many waterlogging hazards, which might in turn trigger large numbers of public reports (Wang, Loo, Zhen, & Xi, 2020). UGC data revealed that there was an uptick in media coverage after the rainfall (Figure 6b), except for a delay between the UGC reporting time and the beginning of the rainfall event, which aligns with the findings of previous studies (Fang, Hu, Shi, & Zhao, 2019). Another reason for the delay might be the policy compliance of media censorship. Despite this, there is an overall high consistency between dynamic UGC reports and the evolution of flooding.

Notably, UGC has several advantages over rainfall data. The number of rainfall observation stations in a city is usually limited and sporadically distributed across a large geographical scope. To a large extent, the spatial and temporal distribution of rainfall computed by spatial interpolation algorithms reflect the difference in precipitation in a whole city and indicate the areas prone to disasters at a regional scale; however, the rainfall data cannot indicate the specific location (e.g. streets, roads) of flooding. In addition, not all rainfall events trigger the occurrence of flooding disasters. Whether the flooding will occur or not depends on the water balance between rainfall and drainage system. For instance, in an area with poor drainage systems, even light rain might result in flooding hazards. Therefore, it is difficult to predict detailed flooding locations (e.g. specific streets and roads) using only rainfall data. In this regard, the geo-tagged disaster names contained in the UGC may compensate for the incompleteness and uncertainty of rainfall data. For instance, from July 6, 2016 to July 14, 2016 (Figure 6b), although rainfall stopped, the disaster continued because the flood water around Nanhu lake had not receded yet. Under the circumstance, the web could still receive the users' observations and reports embedded with flooding locations. Thus, UGC can provide a new approach to track the evolution of disasters.

4.4.2 | Water level

According to the water level data (Figure 7b), the Nanhu Lake water level rose significantly on June 20, July 2 and July 5, and decreased during the two periods (June 21–29 and July 7–17). Along with the increasing water-level trend, the number of UGC reports increased significantly over the three days. After the rainfall stopped (July 6), water was discharged into the Yangtze River, and the water level fell; consequently, the media coverage gradually decreased (Figure 7b). There was a parallel relationship between the changing water level and the number of UGC reports. The disaster information in the UGC and water-level information can be verified using each other.

Notably, there were some discrepancies between the number of UGC reports and the changing tendency of the water level from June 20 to July 2. At this stage, the water level fell gradually, which indicated that the capability of the drainage system in the urban lake district exceeded the runoff caused by rainfall (Figure 7a). Nonetheless, flooded locations were sporadically reported by the media (Figure 7b). This discrepancy might be explained by the potential variations in the hydrological conditions in different districts of the city. Under rainstorm conditions, areas with poor drainage systems remained flooded. Water-gauging facilities were generally not deployed in these areas because such

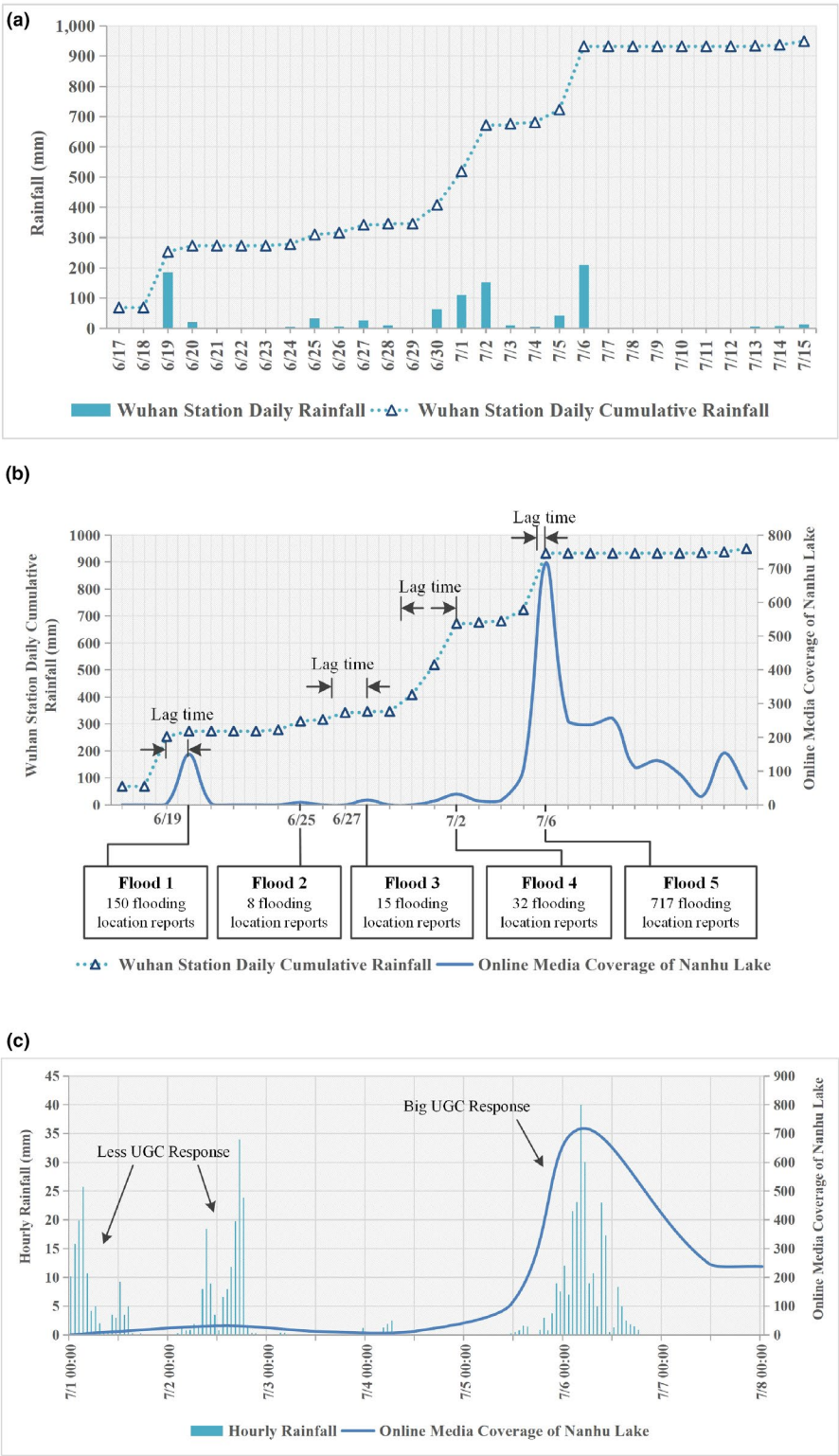


FIGURE 6 The data comparison between rainfall and the coverage of online user-generated content (a) Daily rainfall and accumulated rainfall in the study area; (b) daily rainfall and user-generated content response with flooding locations; and (c) hourly rainfall and user-generated content response with flooding locations

facilities are often built along the watershed (e.g., wetland and riverside). Therefore, in the impervious surface of the inner city without sufficient water-gauging facilities, UGC can supplement flooding information that cannot be recorded by ground-based monitoring facilities.

4.4.3 | Digital elevation model

Water usually flows downhill because of gravity and accumulates in low-lying areas. According to this rule, if the quality of hotspots reported by users is credible, then most hotspots should be located in low-lying areas (Figure 8).

Overall, the flooding locations can be divided into three categories according to the reported frequency: high, medium, and low frequencies. Among the 100 flooding places reported, 71 were located at elevations below 28 m (Table 1). Specifically, in the data set assessed, high-frequency flooding locations with elevations below 28 m (21

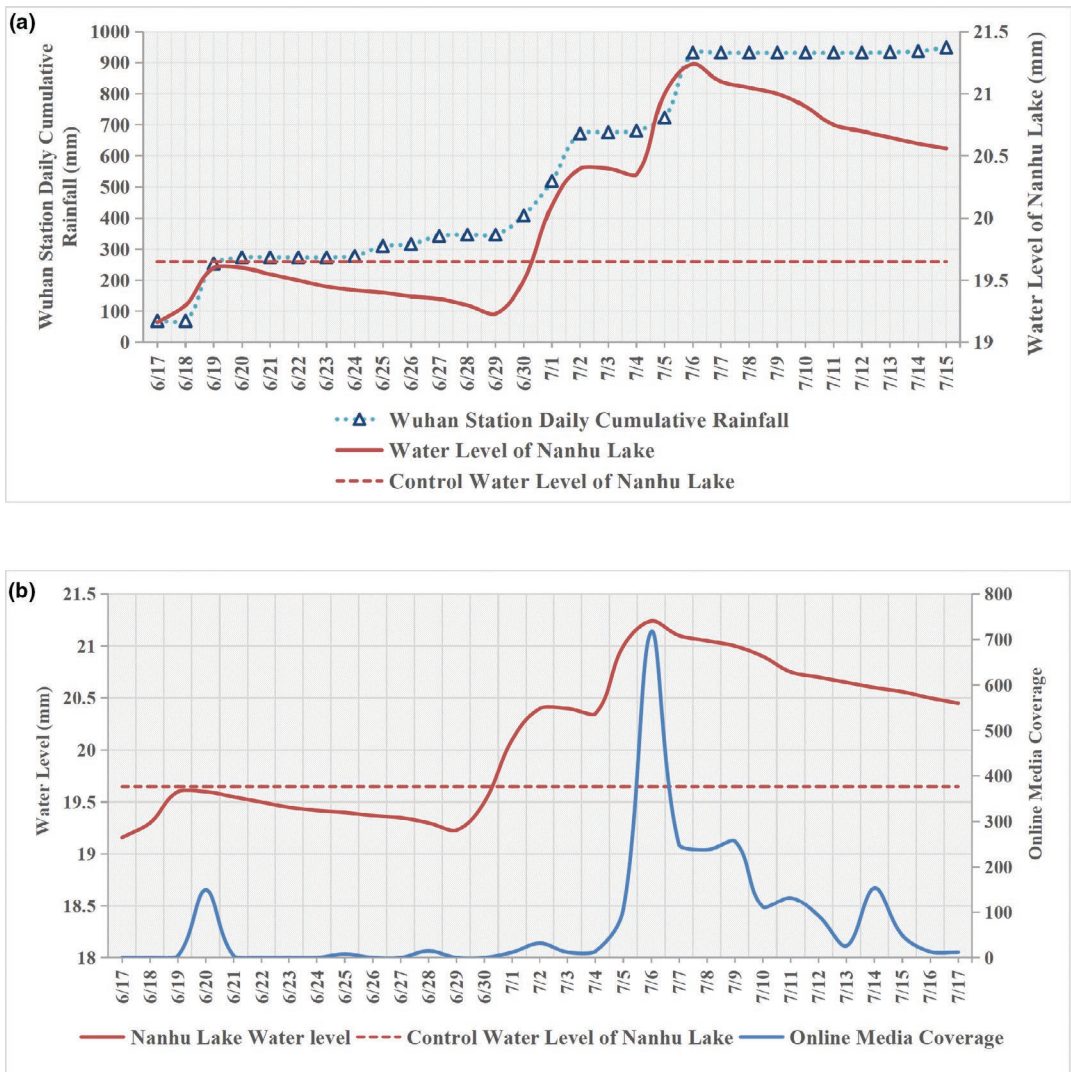


FIGURE 7 The data comparison among rainfall, water level and the coverage of online user-generated content (a) Accumulated rainfall and water-level response; and (b) Water level and user-generated content response

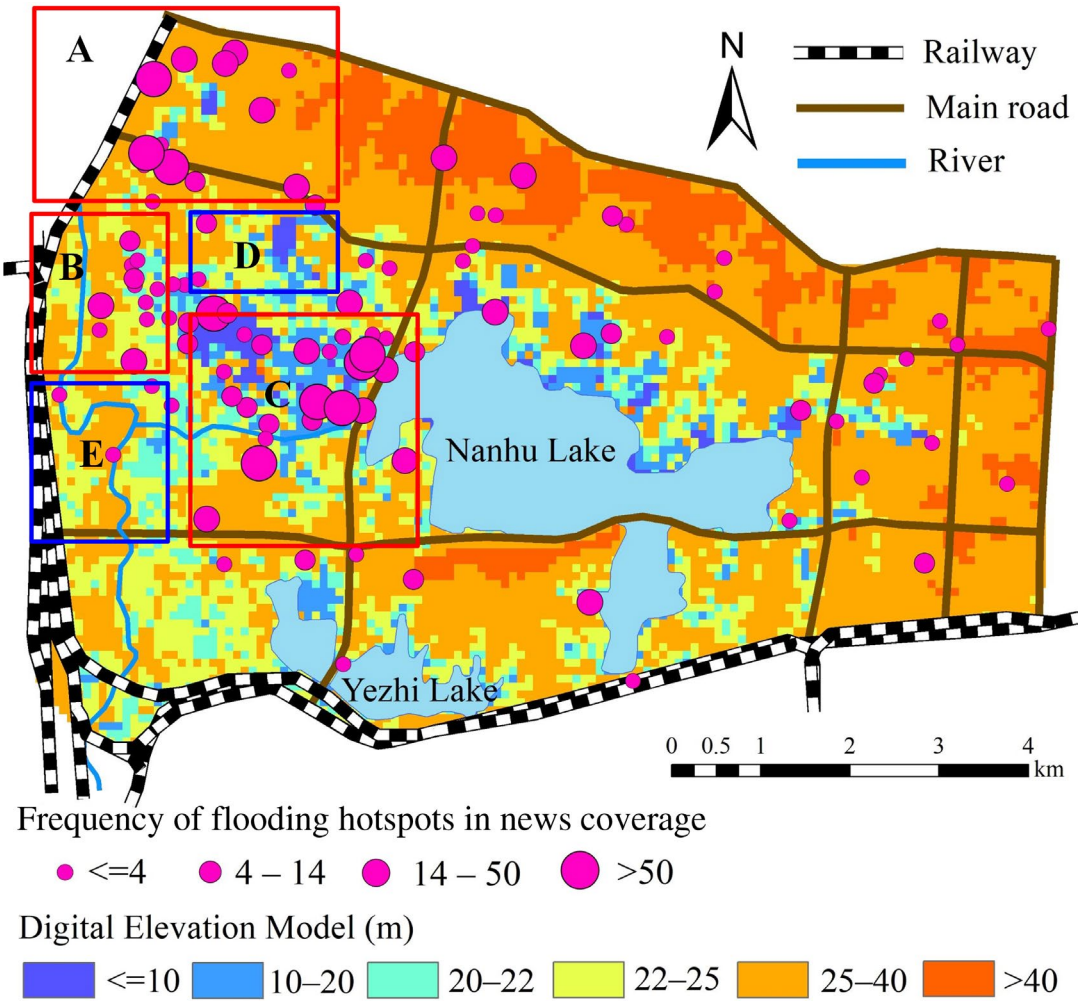


FIGURE 8 Map depicting the reporting frequency and elevation of flooding hotspots

TABLE 1 The number of flooding hotspots varies according to elevation

DEM (m)	High frequency (>10)		Medium frequency (5–10)		Low frequency (<5)		Total numbers of flood points	
	Number	(%)	Number	(%)	Number	(%)	Number	(%)
<28	21	21	13	13	37	37	71	71
28–33	8	8	2	2	7	7	17	17
>33	3	3	1	1	8	8	12	12
Sum	32	32	16	16	52	52	100	

points, 21% of the total flooding hotspots reported) received 67% of media attention (Table 2). The higher the reported frequency in the flooding locations, the lower the elevation. Notably, in our study, none of the locations were higher than 46 m. Although each low-evaluation location would not be reported by UGC, almost all reported names were located in low-lying areas (according to the DEM).

TABLE 2 The reporting frequency for flooding hotspots varies according to elevation

DEM (m)	High frequency (>10)		Medium frequency (5–10)		Low frequency (<5)		Total frequency of flood points	
	Frequency	(%)	Frequency	(%)	Frequency	(%)	Frequency	(%)
<28	1,186	66.7	100	5.6	72	4.0	1,358	76.4
28–33	220	12.4	13	0.7	13	0.7	246	13.8
>33	150	8.4	7	0.4	17	1.0	174	9.8
Sum	1,556	87.5	120	6.7	102	5.7	1,778	

A number of low-lying areas did not receive user observations. For instance, the elevations of parts of zones D and E were lower than 21 m (Figure 8, areas in blue color). According to the water-gauging data, these areas should have been flooded by rainstorms. However, due to the small population in these areas, there were only few disaster reports. From this point of view, the combination of traditional measurement data and DEMs might enable us to gain a better understanding of the disaster situation than that obtained using UGC alone.

5 | DISCUSSION

The collection of flooding location data is the core of disaster emergency management. DHM is an essential tool for revealing the spatial-temporal evolution pattern of disasters, requiring extensive flooding location sources. During the hazard, detailed data that reflect disaster situations play a critical role in effective decision support. Our research results indicate that UGC is of great value as a supplement for disaster risk management.

The changes in accumulative geo-referenced flooding data in the form of webpages responding to the disaster situation are consistent with those of data observed on other platforms (e.g., Twitter and Facebook) (Fohringer, Dransch, Kreibich, & Schröter, 2015). Both data sets have in common that they are produced by non-scientists and citizens. In the internet age, UGC might exist on the web in various forms with great value, and such data sources for DHM should receive more attention from the emergency management sector.

News reports are widely used for risk communication in many countries. User-generated data have unstructured characteristics with low spatial density. Therefore, an efficient tool for retrieving valuable information and making the most of its value is needed. With the support of GeoAI (i.e., WebCrawler, NLP, data mining, and geographic visualization), critical information in UGC can be rapidly mined and accessed for emergency decision-making. Our study provides insights into similar studies on DHM.

Although UGC provides useful data for DHM, its limitations should be noted. For instance, not every disaster will generate as much UGC as our study. Previous studies show that the completeness and accuracy of UGC remain different among various data sources due to a lack of standardization (Senaratne, Mobasheri, Ali, Capineri, & Haklay, 2017). Therefore, a passion for user participation should be fostered. In addition, there might be social and geographical biases in UGC (McLafferty, Schneider, & Abelt, 2020; Su, Lei, Li, Pi, & Cai, 2017). For instance, during the Wuhan disaster, the media coverage was geographically unbalanced on the web (Zeng, Lan, Hamidi, & Zou, 2020). News media would allocate more resources for reports in areas with higher population density or severely damaged areas that might receive more public attention. Therefore, the relationship between news reports and actual disaster situations requires further investigation. The accumulation of more case analyses may deepen our understanding of this issue.

Compared to social media data, which offer access to timely information regarding the situation, the data used are not very suitable for disaster prediction because news media might report the disaster with a temporal delay

(Kitazawa & Hale, 2021): journalists require time to audit the news report before it is released. Thus, it is necessary to integrate social media into flood management to empower disaster prediction and early warnings. Furthermore, with regard to data management, user observation data on the web have many deficiencies. Compared to flood-gauging facilities, which usually have a reliable recoding mechanism, the UGC on the web might disappear as time progresses. Thus, efficient management of UGC must be implemented as well.

Nevertheless, our study shows that UGC has great potential in disaster hotspot tracking, risk communication, and emergency decision-making. At present, the number of internet users is rapidly growing, in line with the quantity of UGC, and UGC is becoming increasingly important, especially when traditional data collection (e.g., monitoring facilities and field investigation) fails and spatial data infrastructure data cannot be updated in a timely manner. As a critical supplementary data source, the combination of UGC and traditional monitoring data can play a bigger role in disaster management. In specific disaster emergency management scenarios, the actual disaster situation can be cross-validated through statistical models, empirical models, and knowledge management methods. Thus, by integrating citizen participation and scientists' efforts, the capability of and confidence in flood disaster management can be increased.

6 | CONCLUSIONS

In summary, web data generated by users during flood disasters are a very promising source for DHM. Therefore, it is necessary to incorporate this source into flood disaster management. Such data can help to better cater to the data source requirements of urban flooding management. The integration of user observations into flooding management (with traditional sources of hydro-information) can improve our response to flood events and better support emergency management.

The look-up table method employed in our study was effective for place name recognition. Notably, many advanced technologies in the areas of artificial intelligence, GIR, and NLP used for text mining and clustering analysis are evolving rapidly. For DHM using UGC, it is necessary to adopt a better method to improve the automation, accuracy, and callback of extracting geographical name entities. Additionally, given the insufficient UGC data source, emerging technologies (e.g., specific platforms, tools, and apps) should be developed for citizens to enable better participation, and incentive policies should be designed to increase citizens' motivation to share disaster reports.

Media reports regarding disasters are not limited to flooding events, and therefore, UGC related to other disasters (e.g., landslides, earthquakes, cyclones, and environmental pollution) should also be mined. Moreover, the concept of the "human as sensor" involves citizen reports, VGI, crowdsourcing data, and citizen science, providing a wider database and exposure than the traditional means. Notably, the potential of UGC (e.g., Twitter and Facebook entries) for DHM needs to be further explored because it may assist research communities to gain a deeper understanding of different UGC data sources applied in DHM and provide insights for effective disaster management.

CONFLICT OF INTEREST

There is no conflict of interest of this study.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

ORCID

Zhongping Zeng  <https://orcid.org/0000-0001-9080-2143>

Qingsong He  <https://orcid.org/0000-0001-9370-3335>

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